

V3 moneyness performance

Same 6 models · 3 companies · 4 regimes · 5-year monthly study · classified by call moneyness $m = S/K$

What this report is

- Companion to V3_5y_monthly_empirical_study. The notebook in this folder is the source of truth; this PDF is written from the same moneyness payload.
- Question: does model ranking change with option moneyness, overall and inside each volatility regime?
- BEST in every table is the lowest RMSE%. That row is bold and green, and the Mark column says BEST. Empty buckets ($n = 0$) show em dashes and have no BEST.

Unchanged experimental design

- Same six models, three companies (SPY, AAPL, MSFT), four V3 regimes, 5-year lookback, monthly recalibration, V3 filters, Euler-corrected Heston step, LSM $n_{\text{paths}} = 2000$ / seed 42.
- Same option contracts as the main study (Monday nearest-ATM listed calls, DTE 7-60, $|S/K - 1| \leq 10\%$). No model-specific re-sampling.
- Pooled $n = 560$ contracts per model (identical keys across models).

Moneyness rule (calls, $m = S / K$)

- DOTM: $m < 0.98$. OTM: $0.98 \leq m < 0.995$. ATM: $0.995 \leq m \leq 1.005$. ITM: $1.005 < m \leq 1.02$. DITM: $m > 1.02$.
- These five bins partition the main-study filter $|m - 1| \leq 10\%$. The Monday nearest-ATM design puts most mass in ATM; DOTM and DITM are sparse and appear mainly in 2008-2009.

Metrics

- $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})^2)}$. Ranking uses this only.
- MAE and Bias \$ in option-premium dollars. $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Early-exercise fraction = mean LSM share of paths that stop before expiry.

Two analyses

- Analysis 1 — Overall moneyness: pool all companies and regimes. Five tables (one per bucket) plus a Model $\times$ Moneyness RMSE% heatmap.
- Analysis 2 — Regime $\times$ moneyness: keep the four regimes separate, pool the three companies. Compact RMSE tables/heatmaps and a winner grid showing whether the best model changes with moneyness inside each regime.

Table M0 · Contract counts (one model = all models)

Regime	DOTM	OTM	ATM	ITM	DITM	Total
2008-2009 Crisis	9	29	32	33	6	109
2013-2014 Normal	0	5	142	3	0	150
2018-2019 Late-cycle	0	4	140	7	0	151
2019-2020 COVID	0	6	139	5	0	150
All regimes	9	44	453	48	6	560

Moneyness rule · American calls, $m = S/K$

Bucket	Rule	n (pooled)	Share
DOTM	$m < 0.98$	9	1.6%
OTM	$0.98 \leq m < 0.995$	44	7.9%
ATM	$0.995 \leq m \leq 1.005$	453	80.9%
ITM	$1.005 < m \leq 1.02$	48	8.6%
DITM	$m > 1.02$	6	1.1%

ATM dominates because the main study picks the nearest-ATM listed call each Monday. DOTM/DITM are not empty because of a new filter — they are rare in that ATM sample, especially after 2009. Every model still sees the same contracts in every bucket.

Page 3 · Analysis 1 · Overall moneyness · Tables M1-M3

Pooled SPY + AAPL + MSFT and all four regimes. Bold green = lowest RMSE%.

Table M1 · DOTM · $m < 0.98$ · $n = 9$ · BEST = GARCH

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	9	34.05	1.4244	-1.1924	-21.63	21.1%	
GARCH	9	21.64	0.8050	-0.0372	+2.38	19.0%	BEST
Heston	9	29.23	0.9680	+0.9132	+24.32	24.5%	
Merton	9	34.64	1.3509	-1.0390	-20.13	21.0%	
GARCH-Merton	9	25.29	0.6120	+0.4969	+14.40	22.8%	
Heston-Merton	9	30.15	1.0959	+1.0959	+26.25	24.1%	

Table M2 · OTM · $0.98 \leq m < 0.995$ · $n = 44$ · BEST = Heston

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	44	49.46	1.2906	+0.1551	+11.45	23.7%	
GARCH	44	48.00	1.1531	+0.6753	+26.97	22.4%	
Heston	44	38.22	1.1941	+0.7458	+24.04	24.9%	BEST
Merton	44	47.77	1.2642	+0.0880	+9.69	21.1%	
GARCH-Merton	44	58.00	1.4045	+1.0952	+39.92	21.2%	
Heston-Merton	44	40.60	1.2858	+0.7081	+24.79	21.5%	

Table M3 · ATM · $0.995 \leq m \leq 1.005$ · $n = 453$ · BEST = Heston

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	453	51.79	1.7660	+0.8007	+29.31	27.2%	
GARCH	453	47.80	1.7973	+1.2940	+31.56	26.8%	
Heston	453	47.34	2.0440	+1.7212	+35.22	28.6%	BEST
Merton	453	50.53	1.7374	+0.7108	+27.59	23.9%	
GARCH-Merton	453	57.91	2.2129	+1.8342	+43.00	24.5%	
Heston-Merton	453	48.23	1.9697	+1.6094	+32.34	23.2%	

Bias \$ = mean(model – market). Bias% = 100 × mean((model – market)/market).

Page 4 · Analysis 1 · Overall moneyness · Tables M4-M5

Pooled SPY + AAPL + MSFT and all four regimes. Bold green = lowest RMSE%.

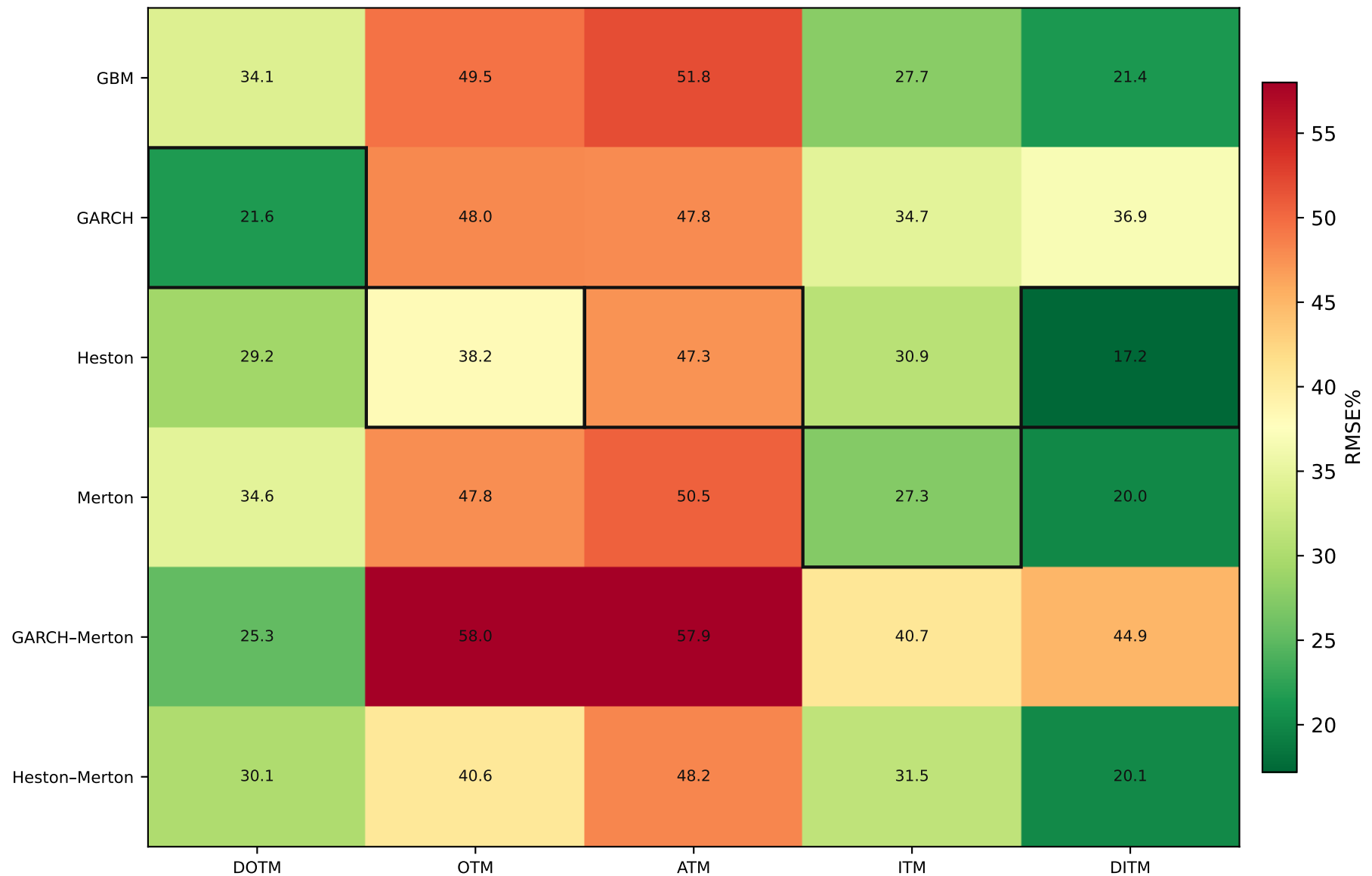
Table M4 · ITM · $1.005 < m \leq 1.02$ · $n = 48$ · BEST = Merton

Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	48	27.69	0.9486	-0.0166	-3.77	30.4%	
GARCH	48	34.67	0.9633	+0.6668	+20.80	28.4%	
Heston	48	30.93	1.2401	+0.9054	+18.06	31.6%	
Merton	48	27.31	0.9035	-0.1262	-6.26	27.3%	BEST
GARCH-Merton	48	40.71	1.1511	+0.9857	+28.26	27.1%	
Heston-Merton	48	31.46	1.2118	+0.8959	+18.76	26.7%	

Table M5 · DITM · $m > 1.02$ · $n = 6$ · BEST = Heston

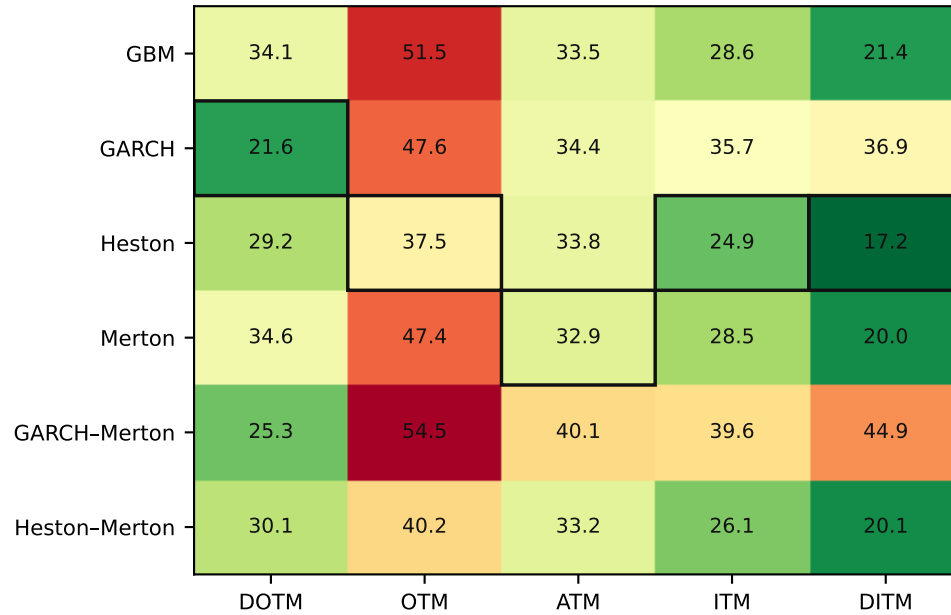
Model	n	RMSE%	MAE	Bias \$	Bias%	Early-ex.	Mark
GBM	6	21.41	0.8883	-0.7720	-8.50	31.0%	
GARCH	6	36.86	0.6065	+0.5236	+24.19	24.2%	
Heston	6	17.20	0.3569	+0.2325	+9.70	31.5%	BEST
Merton	6	20.01	0.8392	-0.7487	-9.35	25.5%	
GARCH-Merton	6	44.94	0.8441	+0.8441	+32.48	21.2%	
Heston-Merton	6	20.11	0.4556	+0.3904	+12.86	33.1%	

ITM/DITM: $S > K$. DOTM/OTM: $S < K$. ATM: $|S/K - 1| \leq 0.5\%$.

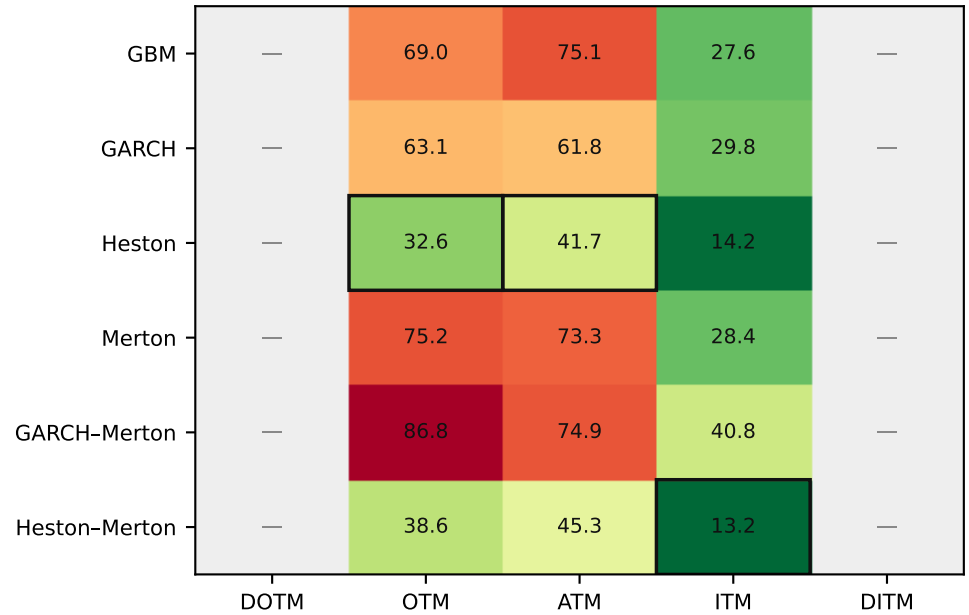


Black outline = lowest RMSE% in that moneyness column. Grey = n = 0. Pooled across companies and regimes.

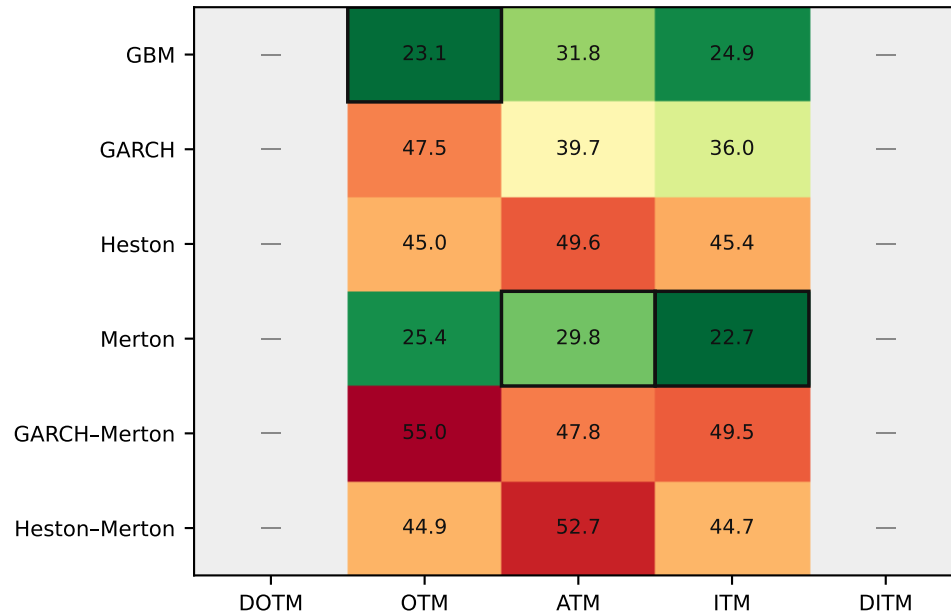
2008-2009 · Crisis



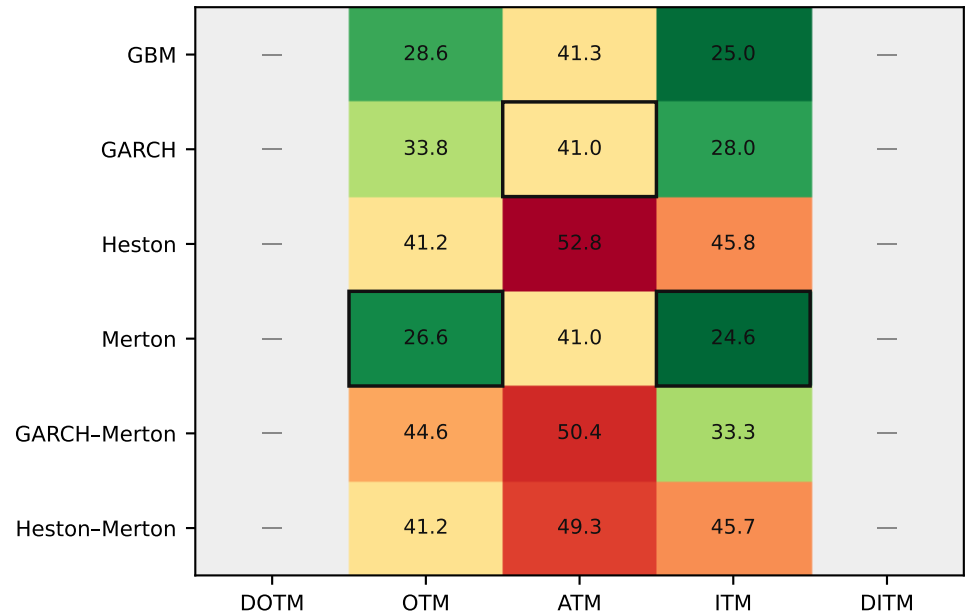
2013-2014 · Normal



2018-2019 · Late-cycle



2019-2020 · COVID



Outlined cell = lowest RMSE% in that regime × moneyiness column. Grey = n = 0.

Table M6 · Best model (lowest RMSE%) by regime × moneyness

Regime	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	GARCH	Heston	Merton	Heston	Heston
2013-2014 Normal	—	Heston	Heston	Heston–Merton	—
2018-2019 Late-cycle	—	GBM	Merton	Merton	—
2019-2020 COVID	—	Merton	GARCH	Merton	—

Table M7 · n contracts by regime × moneyness (shared across models)

Regime	DOTM	OTM	ATM	ITM	DITM
2008-2009 Crisis	9	29	32	33	6
2013-2014 Normal	0	5	142	3	0
2018-2019 Late-cycle	0	4	140	7	0
2019-2020 COVID	0	6	139	5	0

A change in the Table M6 entry across a row means the ranking depends on moneyness inside that regime. Columns with n = 0 cannot change the ranking. Post-2009, ATM holds almost all of the Monday sample.

Does performance change with moneyness?

- DOTM (n = 9): BEST = GARCH with RMSE% = 21.64, Bias% = +2.38.
- OTM (n = 44): BEST = Heston with RMSE% = 38.22, Bias% = +24.04.
- ATM (n = 453): BEST = Heston with RMSE% = 47.34, Bias% = +35.22.
- ITM (n = 48): BEST = Merton with RMSE% = 27.31, Bias% = -6.26.
- DITM (n = 6): BEST = Heston with RMSE% = 17.20, Bias% = +9.70.
- Overall, the ranking does change with moneyness: DOTM → GARCH, OTM → Heston, ATM → Heston, ITM → Merton, DITM → Heston.
- Inside regimes, the BEST model also changes with moneyness in: 2008-2009, 2013-2014, 2018-2019, 2019-2020.
- These comparisons use the Monday nearest-ATM sample, so ATM is the only well-populated bucket after 2009. DOTM/DITM results rest on few 2008–2009 quotes and should be read as descriptive, not as a high-powered test.